# Predictive Interaction Dynamics for Robot-Assisted Laparoscopy with Remote Center-of-Motion and Tissue-Force Constraints

Anonymous Author(s)

***Abstract*—Robot-assisted laparoscopy requires simultaneous tool-tip tracking, remote-center-of-motion (RCM) regulation at the trocar, and limitation of nonlinear tool–tissue contact under actuator constraints. We present a predictive interaction-dynamics controller that combines a constant linear state-transition matrix with a configuration-dependent input map and a joint/task quadratic program whose decision variables are Cartesian tip force and seven auxiliary joint torques. The optimizer imposes horizon-wide linearized RCM, total-command actuator, and adaptive tissue-penetration rows. A dead-zone recursive-least-squares estimator identifies Hunt–Crossley tissue stiffness, while an oscillator-augmented disturbance model supplies an optional periodic free response. Focused regression tests verify that this free response enters the deployed joint/task optimizer and changes its command and adaptive force rows. A separate six-state $H_2/H_\infty$ formulation targets local linear feedback and weighted state/control attenuation; it is distinct from the deployed 20-state constrained controller and does not certify nonlinear tissue force. Closed-loop performance is evaluated in a MuJoCo FR3 protocol with position-impedance, PID-plus-force, and tip-only MPC baselines, matched randomized episodes, configured-delay tests, and RCM/force sensitivity sweeps, all regenerated through the corrected prediction path, including a retuned joint trust-region radius identified by an ablation that isolated it (not the free-response fix) as the cause of an earlier, worse-performing regeneration. The complete joint/task controller reduces nominal maximum RCM deviation to 0.59 mm and tip RMSE to 1.71 mm, versus 5.53 mm RCM deviation and 2.55 mm RMSE for tip-only MPC and 8.13 mm RCM deviation for position impedance. Across eight randomized episodes, joint/task MPC gives $2.49 \pm 0.93$ mm tip RMSE and $0.62 \pm 0.22$ mm maximum RCM deviation, with QP-solve fallbacks in six of eight episodes (11 of $\sim$9600 control steps in aggregate).**

***Index Terms*—Surgical robotics, interaction dynamics, model predictive control, remote center of motion, Hunt–Crossley contact model, $H_\infty$ control.**

## I. Introduction

CONSTRAINED physical interaction can be formulated as an interaction-dynamics problem. The controller shapes how commanded motion, environmental contact, human- or tissue-applied forces, and actuator limits evolve together. Robot-assisted laparoscopy is a demanding instance because three coupled interactions occur at the same time: tool–tissue interaction at the surgical site, tool–trocar interaction at the abdominal wall, and robot–environment interaction through the full manipulator and instrument.

Manuscript received June 2026. Author information omitted for double-anonymous review.

Existing interaction-dynamics shaping methods in surgical robotics commonly use impedance control and admittance-like variants. These methods provide a compliant closed-loop relation between force and motion. The formulation studied here uses a predictive optimization parameterization so that steady-state force offsets, horizon-wide RCM consistency, actuator saturation, and nonlinear tissue penetration can be represented in one local prediction.

We formulate robotic laparoscopy as a constrained interaction-dynamics regulation problem. Rather than predicting the full nonlinear robot dynamics over the horizon, a feedforward layer normalizes the robot into a configuration-invariant linear double-integrator backbone: the state transition of the interaction dynamics is constant, while robot configuration appears only through the input matrix. The present paper lifts this backbone into joint/task coordinates and adds adaptive penetration rows so that tool motion, trocar consistency, tissue interaction, and actuator limits are optimized within one predictive interaction-dynamics framework.

### A. Related Work

*RCM enforcement.* Robot-assisted laparoscopy has historically enforced the RCM *kinematically*, through a constrained-Jacobian formulation projecting the commanded tool-tip motion onto the subspace consistent with a stationary trocar point [11]; more recent schemes move enforcement to the *torque* level, treating the RCM as a holonomic constraint with a Lagrange multiplier and splitting the joint torque into constraint-maintaining and free-motion components via projection-based inverse dynamics [12]. The cited formulations impose the pivot at the current control instant. The present formulation extends this mechanism by imposing a linearized RCM row at each prediction step.

*Predictive variable impedance.* Xue *et al.* [2] regulate interaction dynamics through MPC-selected impedance gains, adaptive Hunt–Crossley estimation, and passivity guarantees; Anand *et al.* [3] use learned dynamics and sampled stiffness optimization. In our terminology, impedance is one realization of interaction dynamics. We also estimate Hunt–Crossley parameters, but use them to form a penetration-output row while optimizing tip force and RCM-authority torque directly. A constraint-matched MPVIC implementation is identified as the next comparative benchmark.

*Robust interaction-dynamics analysis.* The surgical-robot literature often uses passivity to reason about interaction, but

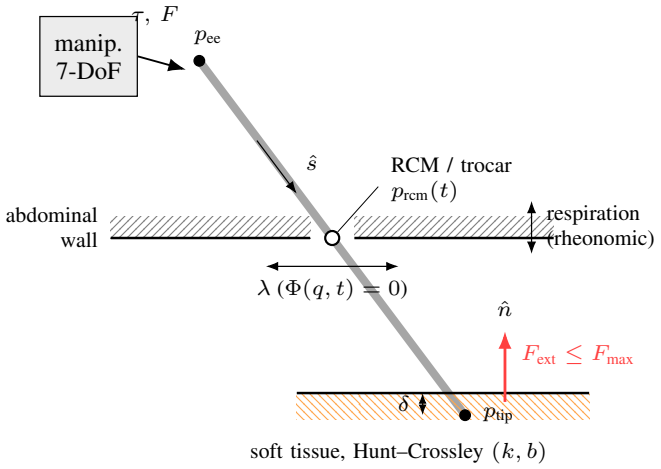



Fig. 1. RCM-constrained laparoscopic plant, mapping the variables of this section onto the physical parts: a 7-DoF manipulator applies joint torque $\tau$ and tip force $F$ (1) through a rigid instrument with flange $p_{ee}$, tip $p_{tip}$, and shaft direction $\hat{s}$. The instrument pivots about the trocar incision—the remote center of motion $p_{rcm}(t)$, which may vary in the general model but is fixed in the experiments—where the lateral deviation $\Phi(q,t)$ (2) must be held at zero against the lateral trocar force $\lambda$. The tip presses on soft tissue modeled by Hunt–Crossley contact (stiffness $k$, damping $b$, penetration $\delta$) (4), producing the contact force $F_{ext}$ regulated relative to the configured force limit $F_{max}$ along the contact normal $\hat{n}$. The implemented respiratory disturbance moves the planar tissue surface along the shaft direction.

passivity is an energy statement and does not by itself bound instantaneous tissue force. We retain an offline $H_2/H_\infty$ study for a six-state linear post-normalization model. Its common-$P$ output is $z = [Q^{1/2}x_e; R^{1/2}F_{\mathrm{mpc}}]$ and therefore certifies weighted state/control performance, not tissue reaction force. Nonlinear contact limitation is evaluated separately through the online penetration rows and simulation stress tests.

The contributions of this paper are: **Predictive interaction-dynamics formulation** (Section III): robotic laparoscopy is posed as an interaction-dynamics optimization problem in which motion accuracy, RCM consistency, tissue interaction, and actuator limits share one local predictive model. The online decision combines Cartesian tip force and seven joint correction torques. **Constant-transition interaction predictor and adaptive interaction constraints**: the predictor has a constant state-transition matrix while robot kinematics and inertia enter through the input map. RCM is represented by structured horizon-wide rows, and dead-zone RLS calibrates a Hunt–Crossley penetration constraint and deep-reference feasibility backstop. **Auditable implementation and evaluation protocol**: exact-ZOH input terms, total feedforward-plus-MPC actuator limits, an oscillator disturbance option, an infeasibility fallback, and configured-delay supervision are integrated around the same interaction-dynamics backbone. Focused tests verify the corrected free-response path, while the closed-loop protocol includes matched baselines, parameter sweeps, randomized episodes, and separate solver-only and controller-call timing.

## II. Interaction Dynamics Formulation

### A. Robot Dynamics

The manipulator equation of motion of [1] is retained:

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) = \tau + J^\top(q)F_{\mathrm{ext}}, \tag{1}$$

where $F_{\mathrm{ext}} \in \mathbb{R}^6$ is now the *tool–tissue* wrench rather than the human-applied wrench. The 7-DoF manipulator carries a 30 cm rigid laparoscopic instrument; the dynamic structure and the operational-space inertia $\Lambda(q) = (J_v M^{-1} J_v^\top)^{-1}$ [8] are unchanged.

### B. Tool–Trocar Interaction Constraint

Let $p_{rcm}(t) \in \mathbb{R}^3$ denote the trocar point, allowed to vary slowly as the patient breathes. With end-effector flange $p_{ee}(q)$, tool tip $p_{tip}(q)$, and shaft direction $\hat{s}(q) = (p_{tip} - p_{ee})/\|p_{tip} - p_{ee}\|$, the perpendicular deviation of the trocar from the tool axis is

$$\Phi(q,t) = \underbrace{\left(I_3 - \hat{s}\hat{s}^\top\right)}_{P_\perp(\hat{s})}\left(p_{rcm} - p_{ee}\right), \tag{2}$$

and the RCM constraint is $\Phi(q,t) = 0$. Although (2) is written as a redundant three-vector, $P_\perp \hat{s} = 0$, so the independent lateral constraint has rank two. Equivalently, one may choose an orthonormal normal-plane basis $E_\perp(\hat{s}) \in \mathbb{R}^{2\times 3}$ and enforce $\phi = E_\perp \Phi \in \mathbb{R}^2$. The implementation keeps the redundant three-component $\Phi$ because $\|\Phi\|_2$ is the physical lateral deviation and componentwise prediction rows are convenient for the QP. Because $p_{rcm}(t)$ depends explicitly on time, the RCM is a rheonomic holonomic constraint. Differentiating the redundant form, $\dot{\Phi} = J_c(q,t)\dot{q} + \partial\Phi/\partial t$ with $J_c = \partial\Phi/\partial q$, and the constrained dynamics may be written as

$$M\ddot{q} + C\dot{q} + G = \tau + J_c^\top \lambda + J^\top F_{\mathrm{ext}}, \tag{3}$$

with $\lambda \in \mathbb{R}^3$ a redundant lateral trocar-force representation satisfying $\hat{s}^\top\lambda = 0$; in a minimal basis this term is $J_\phi^\top \mu$, $J_\phi = E_\perp J_c$, $\mu \in \mathbb{R}^2$. The RCM acts on $q$, not directly on the residual state of the predictive architecture; it is imposed via the kinematic chain $q_k \to (p_{tip,k}, p_{ee,k}) \to \Phi_k$ at every horizon step without affecting the constant-$A_d$ structure.

### C. Tool–Tissue Interaction Constraint

Where [1] modelled the human push as an unknown bounded force absorbed by the integrating disturbance state, the laparoscopic interaction force has known structure. We adopt the Hunt–Crossley model [9], [10]:

$$F_{\mathrm{ext}}(\delta,\dot{\delta}) = k_e\delta^n(1 + b\dot{\delta}), \quad \delta \ge 0, \tag{4}$$

with penetration depth $\delta$, stiffness $k_e$, exponent $n = 1.5$, and damping $b$. Accordingly, $k_e$ has units $\mathrm{N/m^{1.5}}$ in this study. The unknown parameters $\theta = [k_e, b]^\top$ are adapted online (Section IV).

The implementation uses a planar surface with fixed normal $\hat{n} = \hat{s}_0$, so $\delta = \max(0, (p_{tip} - p_c)^\top \hat{n})$ and $\dot{\delta} = \dot{p}_{tip}^\top \hat{n}$. The present validation covers this planar-contact setting, while the equations support extension to a rotating contact frame and curved-surface sliding. Tissue heterogeneity is evaluated through independent episodes at $k_e = 200$, 500, and $1200\,\mathrm{N/m^{1.5}}$.

### D. Interaction Disturbance Model

The lumped disturbance acting on the Cartesian residual dynamics decomposes as

$$d(t) = d_{\text{tissue}}(\delta, \dot{\delta}; \theta) + d_{\text{resp}}(t) + d_{\text{trem}}(t) + d_{\text{tro}}(t) + \epsilon_m(t), \quad (5)$$

where $d_{\text{tissue}}$ is represented explicitly in the constrained output, and $d_{\text{resp}}, d_{\text{tro}}$, and residual model error act through the measured plant. The structured tissue term is predicted by the Hunt–Crossley estimator; the remaining low-frequency residual can be carried by the inherited offset-free disturbance state [5], [15]. Measured stress tests characterize its practical effect, while the online penetration rows govern nonlinear contact limitation.

The important distinction is between state-dependent contact and exogenous uncertainty. Because penetration is a function of the planned tip trajectory, the structured Hunt–Crossley term can be constrained. Friction, sensing error, and unmodelled dynamics cannot be commanded directly; they motivate empirical tightening and stress tests. Force limitation is reported relative to a configured engineering threshold (Section VII).

### E. Kalman Disturbance Augmentation

The offset-free component inherited from [1] is an integrating force-form disturbance state. For the residual error model (13), the augmented predictor is

$$\begin{bmatrix} x_{e,k+1} \\ \hat{d}_{k+1} \end{bmatrix} = \begin{bmatrix} A_d & B_d \\ 0 & I_3 \end{bmatrix} \begin{bmatrix} x_{e,k} \\ \hat{d}_k \end{bmatrix} + \begin{bmatrix} B_d \\ 0 \end{bmatrix} F_{\text{mpc},k} + \begin{bmatrix} 0 \\ w_k \end{bmatrix}. \quad (6)$$

The integrating block is the standard internal model for constant force bias and gives zero steady-state error when the residual is bounded and asymptotically constant, with the usual care needed under actuator saturation [6]. In this paper the Kalman state represents the unstructured remainder after the planned Hunt–Crossley contact has been accounted for: axial trocar friction, sensing bias, and slowly varying model error. The dead-zone tissue estimator therefore improves the offset-free model indirectly by removing the dominant structured component before it is treated as a random walk. In the current v1 code this estimator forms diagnostic residuals and the tip-MPC/free-response branch, while the explicit RLS penetration and RCM rows provide the primary joint/task QP mechanisms.

### F. Internal Model for Periodic Respiration

The random-walk disturbance state (6) predicts future disturbance as constant. That is adequate for slowly varying contact bias and trocar friction, but respiration is quasi-periodic. The controller therefore keeps an oscillator augmentation as an optional internal model:

$$x_{r,k+1} = \begin{bmatrix} \cos\omega_r\Delta t & \sin\omega_r\Delta t \\ -\sin\omega_r\Delta t & \cos\omega_r\Delta t \end{bmatrix} x_{r,k} + w_{r,k}, \quad (7)$$

with $d_{\text{resp},k} = C_r x_{r,k}$ mapped along the insertion direction. Stacking $\xi = [x_e^\top, \hat{d}^\top, x_r^\top]^\top$ gives

$$\xi_{k+1} = \begin{bmatrix} A_d & B_d & B_d C_r \\ 0 & I_3 & 0 \\ 0 & 0 & R(\omega_r\Delta t) \end{bmatrix} \xi_k + \begin{bmatrix} B_d \\ 0 \\ 0 \end{bmatrix} F_{\text{mpc},k} + \begin{bmatrix} 0 \\ w_k \\ w_{r,k} \end{bmatrix}. \quad (8)$$

This augmentation preserves both the QP decision dimension and the constant-$A_d$ residual structure. When coupled into the predicted residual, it changes the horizon free response without changing the decision dimension. The deployed joint/task branch passes this free response into the 20-state QP; focused regression tests confirm that changing the augmented trajectory changes the first optimized command and the adaptive force rows. Its closed-loop benefit is measured directly in Section VI-G: connected to the joint/task QP, disabling the oscillator or detuning its assumed frequency measurably changes tip RMSE, unlike the no-op result of the uncorrected controller.

### G. Prediction-Error Growth over the Horizon

The safety and force rows depend on a finite-horizon prediction. Holding a random-walk disturbance fixed over the horizon is exact for constant disturbances, and the following bound quantifies the approximation for time-varying residuals. If the residual disturbance is Lipschitz, $\|\dot{d}(t)\|_2 \le L_d$, and the current estimation error satisfies $\|d_k - \hat{d}_{k|k}\|_2 \le e_K$, then

$$\|d_{k+i} - \hat{d}_{k+i|k}\|_2 \le e_K + L_d\, i\Delta t, \qquad i = 0, \dots, N-1. \quad (9)$$

Thus the pointwise prediction error grows linearly with horizon depth. Over a finite horizon, the corresponding sequence-energy bound is

$$\left\| w_{k:k+N-1} - \hat{w}_{k:k+N-1|k} \right\|_2 \le \left[ \sum_{i=0}^{N-1} \left( e_K + L_d i \Delta t \right)^2 \right]^{1/2}. \quad (10)$$

A short horizon, receding-horizon re-solving, and an accurate oscillator model reduce this quantity. Equation (10) provides a finite-horizon sequence bound. Extending it to pointwise tissue-force certification calls for a certified contact-force output, a measured residual-energy set, and explicit inter-sample and nonlinear margins.

## III. Predictive Interaction Dynamics Control

The architecture of [1] is inherited in its essential interaction-dynamics structure; this section states what is the same and what is new for laparoscopy. That framework has since been verified on a physical UR5e arm [1]: with a 1.5 kg load hung during a hold, its disturbance-augmented controller stays at 0.023 mm (the encoder noise level), against 1.5 mm without the estimator, over five alternated pairs.

### A. Interaction-Dynamics Normalization

The Layer-1 torque decomposition of [1] is preserved:

$$\tau = \tau_{\text{ff}} + J_v^\top F_{\text{mpc}} + J_\omega^\top F_{\text{orient}} + \bar{N}^\top \tau_{\text{null}}, \quad (11)$$

with $\tau_{\text{ff}} = C\dot{q} + G + J_v^\top \Lambda \ddot{p}_d$. After cancellation the residual error dynamics are

$$\ddot{e} = -\Lambda^{-1}(q) F_{\text{mpc}} + d_{\text{res}}(t), \quad (12)$$

with $d_{\text{res}}$ collecting contact and modelling residuals. This is the configuration-invariant interaction-dynamics backbone of the controller: nonlinear robot dynamics are normalized into

a linear residual model whose state matrix is independent of configuration. Exact zero-order-hold discretization gives

$$x_{e,k+1} = \underbrace{\begin{bmatrix} I_3 & \Delta t I_3 \\ 0 & I_3 \end{bmatrix}}_{A_d \text{ (constant)}} x_{e,k} + \underbrace{\begin{bmatrix} -\frac{1}{2}\Lambda^{-1}(q_k)\Delta t^2 \\ -\Lambda^{-1}(q_k)\Delta t \end{bmatrix}}_{B_d(\rho_k)} F_{\text{mpc},k}, \quad (13)$$

with $x_e = [e^\top, \dot{e}^\top]^\top$. The plant is an LPV system with a *constant* state matrix $A_d$ and a parameter-varying input matrix $B_d(\rho_k)$, the only configuration dependence entering through $\Lambda^{-1}(q_k)$; "constant-$A_d$" is used throughout as shorthand for this structure. Its value is computational: powers of $A_d$ and the cost weights are fixed, while the local input map $B_d(\rho_k)$, the horizon map $\Gamma$, and therefore the numerical Hessian $H = \Gamma^\top \bar{Q}\Gamma + \bar{R}$ are updated for the frozen configuration.

### B. Predictive Optimization with Interaction Constraints

The predictive interaction-dynamics QP extends the tip-force decision with a seven-dimensional auxiliary joint torque,

$$u_k = \begin{bmatrix} F_{\text{mpc},k}^\top & \tau_{a,k}^\top \end{bmatrix}^\top \in \mathbb{R}^{10}. \quad (14)$$

This is the implementation change that gives the optimizer direct authority over the RCM. The prediction state is

$$x_k = \begin{bmatrix} e_k^\top & \dot{e}_k^\top & (q_k - q_{\text{home}})^\top & \dot{q}_k^\top \end{bmatrix}^\top \in \mathbb{R}^{20}, \quad (15)$$

and exact ZOH is applied to both the tip and joint double integrators. With $M^{-1}[J_{\text{tip}}^\top \; I_7]$ mapping $u_k$ to joint acceleration, the joint-position block of $B_d$ contains $\frac{1}{2}\Delta t^2 M^{-1}[J_{\text{tip}}^\top \; I_7]$, while the tip-error block contains its negative projection through $J_{\text{tip}}$. A short horizon $N = 5$ is used to keep the 100 Hz problem small. Stacking the ten-dimensional stage inputs gives $U = [u_k^\top, \ldots, u_{k+N-1}^\top]^\top \in \mathbb{R}^{10N} = \mathbb{R}^{50}$, so $H \in \mathbb{R}^{50\times 50}$ and $h \in \mathbb{R}^{50}$.

$$\min_U \quad \tfrac{1}{2} U^\top H U + h^\top U \quad (16)$$

subject to, for $k = 0, \ldots, N-1$,

$$\|F_{\text{mpc},k}\|_\infty \le F_{\max,u}, \quad (17a)$$

$$\tau_{\min} \le \tau_{\text{ff},k} + \begin{bmatrix} J_{\text{tip},k}^\top & I_7 \end{bmatrix} u_k \le \tau_{\max}, \quad (17b)$$

$$\|\Phi_{\text{lin},k}\|_\infty \le \varepsilon_{\text{rcm}}, \quad (17c)$$

$$\delta_k \le \delta_{\max}(\hat{k}_e, \hat{b}). \quad (17d)$$

where $H = \Gamma^\top \bar{Q}\Gamma + \bar{R}$ and $h = \Gamma^\top \bar{Q}\Phi x_k$. The modifications are:

*1) Effort bounds:* separate the Cartesian correction box from the physical actuator box. The virtual Cartesian correction uses $F_{\max,u} = 8\,\text{N}$. The applied command is $\tau = \tau_{\text{ff}} + J_{\text{tip}}^\top F_{\text{mpc}} + \tau_a$, so the QP constrains this *total* command using the FR3 MuJoCo joint-force ranges: $\pm 87\,\text{Nm}$ for joints 1–4 and $\pm 12\,\text{Nm}$ for joints 5–7. Because the short horizon freezes the current local model, the implemented row uses the current $\tau_{\text{ff}}$ and $J_{\text{tip}}$ for all stages; the final applied torque is clipped to the same box as a guard for fallback or inter-sample held commands.

*2) Predictive RCM rows:* use the analytic RCM Jacobian evaluated at the current configuration,

$$\Phi_{\text{lin},k+i} = \Phi(q_k) + J_c(q_k)(q_{k+i} - q_k). \quad (18)$$

The same rows are imposed at every horizon stage. This is a one-point linearization, not a successive-convexification scheme. The configured band is 0.1 mm; realized nonlinear deviation must therefore be evaluated separately under curvature, model freezing, actuator limits, and the 10 ms hold. A per-stage joint trust-region row, $\|q_{k+i} - q_k\|_\infty \le r_q$, directly enforces the deviation bound that Lemma 1 assumes, rather than leaving it unconstrained; $r_q$ is a configured constant, not a value derived from $L_\Phi$, so this row bounds the linearization's operating region without certifying its curvature term. An initial $r_q = 10\,\text{mrad}$ sat at a fragile QP-feasibility boundary for this benchmark's required joint motion: sweeping $r_q$ showed non-monotonic solver behavior (intermittent inaccurate solves and fallbacks) below about 18 mrad and clean, stable behavior at and above it. The reported results use $r_q = 20\,\text{mrad}$, comfortably inside the stable range.

**Lemma 1** (RCM linearization error)**.** *Let $\Phi(\cdot, t_k)$ be twice continuously differentiable on the operating region, with curvature bound $L_\Phi := \sup_q \|\partial^2\Phi/\partial q^2\|_2$, and let the implemented linearization $\Phi_{\text{lin}}$ be taken about the current configuration $q_k$. If the predicted trajectory satisfies $\|q_{k+i} - q_k\|_2 \le v_q\, i\, \Delta t$, with $v_q$ a bound on the predicted joint speed over the horizon, then*

$$\left\|\Phi(q_{k+i}, t_k) - \Phi_{\text{lin}}(q_{k+i}, t_k)\right\|_2 \;\le\; \tfrac{1}{2}\, L_\Phi\, (v_q\, i\, \Delta t)^2. \quad (19)$$

*Proof:* Taylor's theorem with second-order Lagrange remainder, applied componentwise to $\Phi$ about $q_k$: the constant and first-order terms are $\Phi_{\text{lin}}$ and the remainder of each component is bounded by $\frac{1}{2}L_\Phi\|q_{k+i} - q_k\|_2^2$; substituting the trajectory bound gives (19). ∎

Lemma 1 identifies one source of the realized-versus-linearized RCM gap. The evaluation protocol includes an RCM-band sweep; an online $L_\Phi$ estimate and a certified curvature back-off (as opposed to the fixed trust-region constant above) are outside the present implementation.

*3) Adaptive penetration rows:* convert the force threshold into a position-output constraint. Since

$$\delta_k = (p_{d,k} - p_{\text{surf},k})^\top \hat{n} - \hat{n}^\top e_k, \quad (20)$$

the row is affine in the predicted state. With empirical tightening $\rho_F = 0.33$,

$$\delta_{\max} = \left[\frac{\rho_F F_{\max}}{\hat{k}_e(1 + \hat{b}\max(\dot{\delta}, 0))}\right]^{1/n}. \quad (21)$$

The implementation uses $\rho_F = 0.33$ as a provisional empirical margin for the configured 3 N engineering threshold. At $k_e = 1200\,\text{N/m}^{1.5}$, varying $\rho_F$ from 0.25 to 0.33 to 0.40 changes the measured peak from 2.37 to 2.85 to 3.23 N; the coefficient is task- and model-dependent and requires recalibration before transfer to a different contact model or procedure.

*4) Terminal treatment and infeasibility:* use a scaled terminal state penalty. If the joint/task QP does not return a solved status, the controller uses tip MPC plus soft null-space RCM regulation for that update instead of applying zero feedback. This fallback preserves active feedback but relaxes the hard RCM rows. It is not a recursive-feasibility or stability certificate; such a certificate would require a robust invariant terminal set and an online terminal constraint [13].

*Prediction chain.* Unlike tip-only MPC, this formulation carries joint position and velocity as state variables, so $q_{k+i}$ enters (17c) directly. No pseudoinverse reconstruction or external redundant increment is needed.

*Feasibility backstop.* A reference may be initialized centimetres inside tissue, making the first local QP infeasible before the receding horizon can retreat. The implementation first projects such a reference to the same estimated penetration set (21). The horizon rows remain active; the projection is a fail-safe feasibility backstop, not the sole force-limiting mechanism.

### C. Impedance as the Unconstrained Special Case

The impedance-equivalence theorem of [1] carries over unchanged when the constraints (17c)–(17d) are inactive: the unconstrained predictive interaction-dynamics optimizer is an affine state feedback $F_{\text{mpc}} = K_{\text{eff}} e + D_{\text{eff}} \dot{e}$ and the closed loop is the classical task-space impedance [7]

$$\Lambda(q)\ddot{e} + D_{\text{eff}}\dot{e} + K_{\text{eff}} e = -F_{\text{ext}}, \tag{22}$$

with $(K_{\text{eff}}, D_{\text{eff}})$ the unconstrained LQR gains. Impedance is therefore not the main design object; it is the unconstrained realization of the same interaction-dynamics regulator. Prediction adds value where the controller must depart from this static gain—at the RCM, force, or effort constraint boundary, and through the disturbance-augmented free response.

## IV. Adaptive Tissue Estimator

The Hunt–Crossley parameters $\theta = [k_e, b]^\top$ are unknown and patient-specific. The contact law (4) is not linear in $\theta$ but is linear in the lumped vector $\beta = [k_e,\ k_e b]^\top$, since $F_{\text{ext}} = \phi^\top \beta$ with regressor $\phi = [\delta^n,\ \delta^n \dot{\delta}]^\top$. A dead-zone recursive-least-squares law runs on $\beta$, recovering $k_e = \hat{\beta}_1$ and $b = \hat{\beta}_2/\hat{\beta}_1$. Define the implementation gate

$$g_k = \mathbb{1}\{\delta_k > \delta_{\text{PE}} \wedge |r_k| > \delta_0\}, \quad r_k = F_{\text{ext},k}^{\text{meas}} - \phi_k^\top \hat{\beta}_k, \tag{23}$$

and $L_k = \Gamma_k \phi_k / (\alpha + \phi_k^\top \Gamma_k \phi_k)$. The parameter and covariance updates are both gated:

$$\hat{\beta}_{k+1} = \begin{cases} \hat{\beta}_k + L_k r_k, & g_k = 1, \\ \hat{\beta}_k, & g_k = 0, \end{cases} \tag{24}$$

$$\Gamma_{k+1} = \begin{cases} \alpha^{-1}(\Gamma_k - L_k \phi_k^\top \Gamma_k), & g_k = 1, \\ \Gamma_k, & g_k = 0. \end{cases} \tag{25}$$

Here $\delta_{\text{PE}} = 1\,\text{mm}$, $\delta_0 = 0.05\,\text{N}$, and $\alpha = 0.995$. Projection keeps the recovered $\hat{\theta}_k$ in a compact box $\Theta$, bounding the parameter estimate.

*Covariance windup and loss of excitation.* Long static holds and free-space motion drive the regressor energy toward zero; naive forgetting can then inflate $\Gamma_k$ and move $\hat{\theta}$ on noise. The implementation uses three concrete safeguards: updates are disabled below 1 mm penetration, residuals inside a 0.05 N dead zone are ignored, and covariance eigenvalues are clipped to a finite interval. Projection keeps $(\hat{k}_e, \hat{b})$ in the configured parameter box. Test T1 verifies zero no-contact drift and bounded covariance.

The estimator runs at 100 Hz and its sole deployed role is to calibrate (21). The regenerated stiffness sweep gives final errors of 4.9%, 4.3%, and 6.6% for true $k_e = 200$, 500, and $1200\,\text{N/m}^{1.5}$, respectively.

## V. Offline $H_2/H_\infty$ Linear Design

### A. Linear Interaction Plant Used for Synthesis

Interaction-dynamics normalization leaves a linear residual plant at each frozen configuration:

$$x_{e,k+1} = A_d x_{e,k} + B_d(\rho_k)\big(F_{\text{mpc},k} + w_k\big), \tag{26}$$

where $w_k$ is the force-form residual after subtracting the structured Hunt–Crossley prediction. This is the six-state tip model for which the offline controller is synthesized. The common-$P$ script defines the performance output as

$$z_k = \begin{bmatrix} Q^{1/2} x_{e,k} \\ R^{1/2} F_{\text{mpc},k} \end{bmatrix} = C_z x_{e,k} + D_z F_{\text{mpc},k}, \tag{27}$$

so $\|T_{zw}\|_\infty$ bounds a weighted state/control-output channel. Its lower block is the commanded corrective force $F_{\text{mpc}}$, which scopes the certificate to weighted state/control performance. Tissue reaction is governed separately by the nonlinear penetration row (17d) and evaluated in the simulation stress tests.

### B. State-Feedback $H_2$ and $H_\infty$ Synthesis

The $H_2$ design is the discrete LQR solution for the six-state, three-input tip model. This is distinct from passivity-preserving fixed-impedance $H_\infty$ synthesis [4]. When the same tip model and DARE terminal cost $P_2$ are used in an unconstrained finite-horizon problem, its first move equals the infinite-horizon LQR/impedance gain. The offline verification gives the expected gain equality to numerical tolerance. This check validates the six-state design implementation when run with the same nominal plant data; it does not establish gain equivalence for the deployed 20-state, ten-input joint/task QP.

For the robust design, a common-Lyapunov discrete bounded-real LMI is solved over the operational-space inertia polytope [17]. The LMI is written with the signed feedback $F_{\text{mpc}} = \tilde{K} x_e$, where $\tilde{K} = -K$ under the Riccati convention $F_{\text{mpc}} = -K x_e$. With $X = P^{-1} \succ 0$ and $W = \tilde{K} X$, the per-vertex condition is

$$\begin{bmatrix} -X & (A_d X + B_d W)^\top & 0 & (C_z X + D_z W)^\top \\ A_d X + B_d W & -X & B_d & 0 \\ 0 & B_d^\top & -\gamma I & 0 \\ C_z X + D_z W & 0 & 0 & -\gamma I \end{bmatrix} \prec 0. \tag{28}$$

Solving one $(X, W, \gamma)$ for all vertices gives a single storage function $V = x^\top P x$ and one signed feedback gain $\tilde{K} = W X^{-1}$. The v1 script `lap_commonP_sdp.py` uses the same exact-ZOH $B_d = [-\frac{1}{2}\Lambda^{-1}\Delta t^2; -\Lambda^{-1}\Delta t]$ as the

controller and independently checks LMI residuals, closed-loop spectral radii, and frequency-sweep norms; Section VI-F reports the certified values against the regenerated nominal FR3 plant.

### C. Relation to the Deployed Joint/Task QP

The reported controller uses a 20-state prediction, a ten-input decision, joint-state weights, auxiliary-torque weights, and a scalar terminal multiplier. The six-state gains and storage matrices are evaluated as a separate local design study, while the deployed QP uses its full joint/task model and cost. Full-model synthesis and gain matching provide a direct route for connecting these two analysis levels.

### D. Scope of the Common-$P$ Certificate

The LMI certifies the linear induced-gain statement

$$\|z\|_2 \le \gamma \|w\|_2, \qquad \|z\|_\infty \le \|z\|_2. \tag{29}$$

Equation (29) is therefore used only to define the linear weighted-performance analysis. Nonlinear contact behavior is evaluated by the separate penetration-constraint protocol, and no pointwise force-invariance or recursive-feasibility claim is made.

## VI. Simulation Study

### A. Setup

The architecture is implemented in MuJoCo 3.6.0 on a 7-DoF Franka FR3 with a 0.30 m analytical instrument. The trocar is initialized on the home-pose shaft at $p_{\text{rcm}} = [0.307, 0, 0.440]^\top$ m, with lateral stiffness $K_w = 200$ N/m and axial Coulomb–viscous friction. The nominal Hunt–Crossley surface uses $k_e = 500\,\text{N/m}^{1.5}$, $n = 1.5$, $b = 0.1$ s/m, and 3 mm respiratory motion at 0.2 Hz. The task approaches the surface, commands 15 mm penetration, holds, and retracts. The reported disturbance set comprises the respiratory displacement and a 5 N lateral load from 5–7 s. The online QPs are solved with OSQP [18]. The plant runs at 1 kHz and the nominal $N = 5$ benchmark QP at 100 Hz. The latency stress test separately requests a solve every plant step and activates the soft-RCM tip-MPC fallback when configured actuator delay exceeds 4 ms.

The complete implementation contains Layer-1 bias/feedforward compensation, exact-ZOH joint/task prediction, total-command actuator boxes, decision-coupled RCM rows, adaptive penetration rows, deep-reference projection, guarded RLS, the optional oscillator disturbance model, and a tip-MPC/soft-RCM fallback for infeasible or high-delay updates. The $H_2/H_\infty$ gains and common-$P$ certificate are computed offline in `lap_hinf_design.py` and `lap_commonP_sdp.py`. They are not loaded by the constrained QP.

TABLE I
Measured Results — Palpation Through a Compliant Trocar (Respiring Tissue + 5 N Lateral RCM Load)

| Controller | Tip RMSE (mm) | SS err. (mm) | RCM dev. (mm) | Resp. res. (mm) | Peak $F$ (N) |
|---|---|---|---|---|---|
| Position impedance | 8.64 | 3.94 | 8.13 | 2.29 | 0.85 |
| PID + force governor | 8.69 | 4.91 | 8.13 | 2.08 | **0.78** |
| Tip-only MPC | 2.55 | **0.64** | 5.53 | 0.56 | 1.19 |
| **Joint/task MPC** | **1.71** | 1.41 | **0.59** | **0.37** | 1.19 |

### B. Evidence Status and Regeneration Protocol

The numbers reported below were regenerated from the corrected controller (the free-response fix of Eq. (8), the joint trust-region row of Section III, the 4 ms configured-delay threshold, and the piecewise-gated RLS update of Eqs. (24)–(25)), using the external FR3 bridge identified by `PHRI_SIM_DIR`. Of these, an ablation attributes most of the headline-table shift to the trust-region row rather than to the free-response fix (Section VI); all four changes are active in every number reported below. Any earlier archived artifact that predates this correction is superseded by the values below; none is used in this paper.

The evaluation reruns four matched controllers—position impedance, PID plus force governor, tip-only MPC, and joint/task MPC—on the same nominal episode, then executes the stiffness, force-threshold, force-tightening, RCM-band, configured-delay, trocar-clearance, adverse-sensing, and oscillator-detuning studies. Eight fixed-seed episodes retain raw metrics, 95% confidence intervals, extrema, solver statuses, fallback counts, torque saturation, and force-threshold exceedances. Timing is split into OSQP setup/update/solve time and complete controller-call time.

Six focused tests run without the external bridge. They verify actionable dependency failure; forwarding of the augmented free response to the joint/task QP; sensitivity of the first optimized command and adaptive force rows to that response; preservation of primary-failure status and aggregate solver time during fallback; and rejection of an unconverged $H_\infty$ Riccati iteration. These are software regression checks rather than closed-loop performance validation.

### C. Tracking and Disturbance Rejection

Table I and Fig. 2 compare position impedance, PID-plus-force regulation, tip-only MPC, and the complete constrained joint/task MPC on the same 12 s palpation episode.

The results show the same tradeoff structure as the pre-correction controller. Position impedance and PID-plus-force have lower contact peaks because they under-track the insertion command: their tip RMSE is about 8.6–8.7 mm and their RCM excursion exceeds 8 mm. The force governor further reduces peak force from 0.85 to 0.78 N but increases steady-state tracking error. The joint/task controller improves on tip-only MPC in both tip RMSE (1.71 versus 2.55 mm) and maximum RCM deviation (0.59 mm, an order of magnitude below tip-only MPC's 5.53 mm and the impedance baselines' 8.13 mm), at the cost of higher steady-state error (1.41 versus

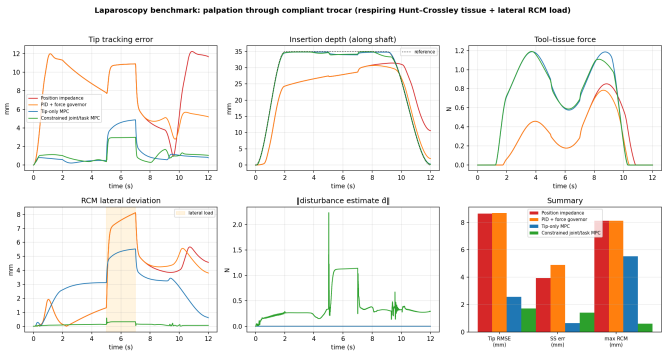



Fig. 2. Palpation benchmark through a compliant, respiring trocar: tip tracking error, insertion depth, tool–tissue force, RCM lateral deviation, diagnostic residual estimate, and summary metrics for the four controllers of Table I.

0.64 mm). This uses the retuned joint trust-region radius of Section III: an initial 10 mrad value sat at a fragile QP-feasibility boundary and inflated both RCM deviation and tip RMSE on this same benchmark (0.82 and 2.61 mm) before retuning to 20 mrad; a separate ablation there also showed the free-response fix itself, in isolation, changes this benchmark's numbers only marginally, so the trust-region value—not the free-response fix—is responsible for that earlier regression. The evidence supports decision-coupled RCM authority as the controller's mechanism; §VII returns to this framing.

### D. Randomized Repeated Trials

Eight fixed-seed episodes vary tissue stiffness (300–900 $\mathrm{N/m^{1.5}}$), Hunt–Crossley damping, respiration amplitude and frequency, sensor noise and bias, and lateral load (3–7 N). All controllers receive the same realization for each seed. Figure 3 reports mean and sample standard deviation. With the retuned trust-region radius, joint/task MPC gives $2.49 \pm 0.93$ mm tip RMSE, $0.62 \pm 0.22$ mm maximum RCM deviation (episode maximum 0.90 mm, 95% CI $[0.43, 0.80]$ mm), and $1.38 \pm 0.28$ N peak tissue force. QP-solve fallbacks occur in six of the eight episodes (0–3 per episode, 11 of ∼9600 control steps in aggregate, down from 34 before retuning) and six of eight episodes exceed a 0.5 mm RCM threshold; no fallback approaches divergence, and no step saturates torque. In comparison, PID-plus-force gives $9.57 \pm 1.17$ mm RMSE and $8.112 \pm 1.589$ mm RCM deviation, position impedance gives $8.79 \pm 0.68$ mm RMSE and $8.125 \pm 1.601$ mm RCM deviation, and tip-only MPC gives $2.60 \pm 0.57$ mm RMSE and $5.557 \pm 0.582$ mm RCM deviation, all with zero recorded solver fallbacks. Joint/task MPC's tip RMSE remains somewhat higher and more variable than in the nominal episode, reflecting the same tracking-versus-RCM tradeoff across a wider operating range, though both the mean and the fallback rate are markedly lower than under the mistuned trust radius.

### E. Verification of the New Mechanisms

Four targeted checks (Fig. 4) confirm the additions.

*(T1) Adaptive RLS.* At nominal stiffness the estimate converges to $\hat{k}_e = 478.4\,\mathrm{N/m^{1.5}}$ (4.3% error); during no contact the estimate has zero drift and covariance growth is $\times 1.0$. The

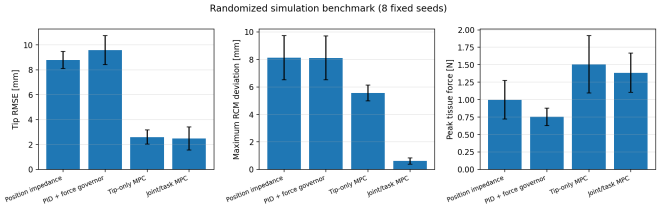



Fig. 3. Eight randomized episodes with matched realizations across controllers. Bars show means and error bars show sample standard deviations.

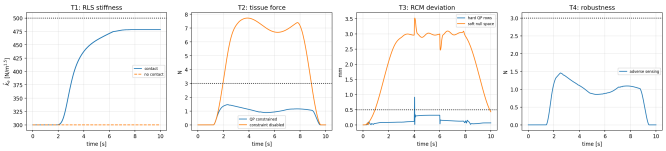



Fig. 4. Verification tests T1–T4: adaptive RLS and no-contact gating; force limitation under a deep reference; decision-coupled versus soft RCM regulation; and adverse sensing with stick–slip friction.

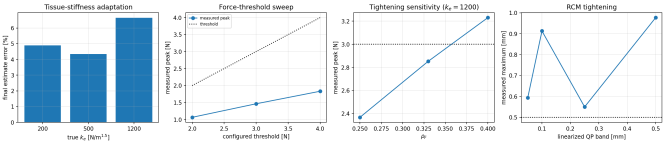



Fig. 5. Multi-condition verification: tissue-stiffness estimation, configured force-threshold sweep, empirical $\rho_F$ sensitivity, and RCM-band tightening.

wider sweep gives 4.9%, 4.3%, and 6.6% final errors at true stiffness 200, 500, and 1200 $\mathrm{N/m^{1.5}}$.

*(T2) Deep-reference feasibility stress test.* The main benchmark uses a 15 mm command. A separate conservative 6 cm in-tissue reference tests recovery from an initially infeasible command: the QP rows and feasibility backstop hold the peak at 1.46 N under a configured 3 N threshold, versus 7.72 N when the limit is set to 100 N.

*(T3) RCM authority.* This targeted mechanism test compares the same MPC tracking law with either decision-coupled RCM rows or a soft null-space RCM term; it is distinct from the position-impedance row in Table I. Under the 5 N lateral load, the two variants give 0.913 and 3.522 mm maximum deviation, respectively. Sweeping the configured linearized band from 0.05 to 0.50 mm gives measured maxima of 0.591, 0.913, 0.550, and 0.977 mm — a non-monotonic pattern under the corrected controller, unlike the smoother trend measured before the free-response fix. This sweep does not identify a best-performing band; the deployed 0.1 mm configuration sits in the middle of the tested range rather than at either extreme. These empirical results characterize the implemented linearized constraint; online evaluation of $L_\Phi$ will extend them to a certified nonlinear curvature margin.

*(T4) Sensor robustness.* Under the analytical sensing model, Gaussian force noise ($\sigma_F = 0.1$ N), 0.05 N bias, and 4 N stick–slip breakaway force give a 1.46 N peak and $\hat{k}_e = 515.0\,\mathrm{N/m^{1.5}}$. The dead zone and projection prevent estimator divergence.

TABLE II
RESPIRATION-OSCILLATOR SENSITIVITY

| Estimator | no osc. | osc. $-20\%$ | matched | osc. $+20\%$ |
|---|---|---|---|---|
| Resp. residual (mm) | 0.506 | 0.487 | 0.490 | 0.507 |
| Tip RMSE (mm) | 1.709 | 1.739 | 1.707 | 1.723 |

### F. Offline $H_2/H_\infty$ Design, Computed

The linear design of Section V is solved by two reproducibility scripts. First, `lap_hinf_design.py` verifies the nominal six-state $H_2$ result: the closed-loop spectral radius is 0.8182, and the unconstrained tip-MPC first-move gain with terminal cost $P_2$ matches the infinite-horizon LQR gain to $9.78 \times 10^{-10}$. The same script gives a local $H_\infty$ minimum $\gamma = 1.0081$; because that minimum is marginally stabilizing, the reported comparison design uses a $1.5\times$ margin, $\gamma = 1.5122$, with spectral radius 0.9728. Second, `lap_commonP_sdp.py` solves the common-storage LMI (28). It certifies $\gamma = 1.480$ over inertia scales $\{0.5, 1.0, 2.0\}$, with strict residual $-4.4 \times 10^{-8}$, maximum closed-loop spectral radius 0.9936, and direct frequency-sweep norm 1.1083. These numbers apply to the weighted state/control output of the offline tip model, not to tissue force or the deployed joint/task QP.

### G. Robustness Stress Tests

Three simulation stress tests probe oscillator sensitivity, configured actuator delay, and trocar-model sensitivity, all rerun through the corrected free-response path.

*A. Respiration oscillator.* With the free response now reaching the joint/task QP, disabling the oscillator or detuning its assumed frequency by $\pm 20\%$ measurably changes closed-loop behavior, unlike the identical-to-three-decimals result of the uncorrected controller (Table II). Matching the assumed 0.25 Hz frequency to the true rate gives 1.707 mm tip RMSE; disabling the oscillator gives 1.709 mm; detuning by $-20\%$ gives 1.739 mm and by $+20\%$ gives 1.723 mm. With the retuned trust region, the QP has more room to compensate for an inaccurate internal disturbance model, so the spread across all four conditions is small ($\leq 2\%$ of the mean) and the matched frequency gives, if anything, a marginal edge; the earlier trust-region value overstated the oscillator's apparent sensitivity by leaving the QP less free to correct for a mismatched model. The command-sensitivity regression tests verify structurally that the oscillator setting has a genuine, if now small, effect.

*B. Configured actuator delay.* A supervisor requests a QP update every 1 ms while a command buffer inserts the configured delay. It retains constrained joint/task control only at 0 ms and switches above the 4 ms configured threshold to tip MPC with soft null-space RCM regulation (Table III); with the corrected controller, the 5 ms case now falls back as well. The experiment characterizes configured actuator-delay sensitivity; computation and communication timing are reported separately.

All configured-delay runs remain stable in simulation. Under the corrected controller, only the 0 ms case retains constrained joint/task control (0.32 mm RCM deviation); the 4 ms threshold now puts every tested nonzero delay, including the 5 ms case, into the soft-RCM fallback, where RCM deviation jumps to 5.6–6.0 mm. This is a narrower stable-constrained window than the pre-correction measurement suggested. The switch serves as a simulated stability backstop, with measured deadline supervision and delay-aware constrained MPC forming the next implementation stage.

TABLE III
CONFIGURED ACTUATOR-DELAY SENSITIVITY

| Configured delay | 0 ms | 5 ms | 10 ms | 20 ms |
|---|---|---|---|---|
| Tip RMSE (mm) | 1.61 | 2.78 | 2.79 | 3.25 |
| Peak tip error (mm) | 3.42 | 5.81 | 5.81 | 5.91 |
| Max RCM dev. (mm) | 0.32 | 5.67 | 5.67 | 6.04 |
| Control branch | joint/task | fallback | fallback | fallback |

TABLE IV
TROCAR DEADZONE AND FRICTION

| Radial clearance | 0 mm | 1 mm | 2 mm |
|---|---|---|---|
| Max RCM dev. (mm) | 0.60 | 0.82 | 0.82 |
| SS tip error (mm) | 0.36 | 0.36 | 0.36 |
| Peak $F$ (N) | 1.19 | 1.19 | 1.19 |

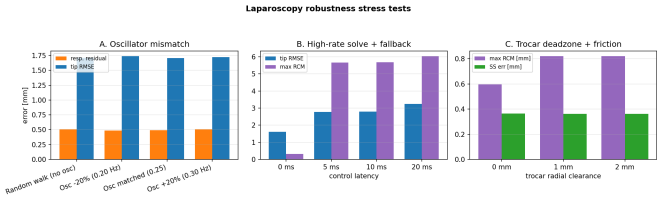



Fig. 6. Robustness stress tests: oscillator/frequency mismatch (A), configured actuator delay with fallback (B), and trocar deadzone with stick–slip friction (C).

*C. Trocar deadzone and friction.* Adding a radial port clearance to the compliant trocar, with axial Coulomb–viscous friction always present (Table IV).

Unlike the pre-correction measurement, the corrected controller shows a modest but real sensitivity to radial clearance in RCM deviation: it rises from 0.60 to 0.82 mm between 0 and 1 mm clearance, with no further change from 1 to 2 mm. Steady-state tip error and peak tissue force are essentially unaffected by clearance. Geometric backlash and deformable-trocar dynamics define the next port-model evaluation.

## VII. DISCUSSION

*Positioning.* The pHRI architecture [1] supplies the interaction-dynamics normalization, constant-$A_d$ tip model, and impedance interpretation. The laparoscopic contribution is the implemented joint/task lifting that puts tool–trocar and tool–tissue interaction constraints in the same online optimization. The paper has two distinct kinds of evidence: an offline weighted-performance certificate for a six-state tip model and measured nonlinear simulation evidence for the deployed 20-state constrained QP.

*Prediction scope.* Lemma 1 identifies the curvature term between linearized and realized RCM motion, while (10) converts pointwise disturbance growth into a finite-horizon

sequence bound. These results describe local prediction error; they do not supply a nonlinear RCM margin, robust terminal set, or pointwise tissue-force invariant.

*Operating envelope and evidence scope.* The implemented study uses a planar analytical Hunt–Crossley surface with fixed normal, an analytical trocar and force sensor, local one-point RCM linearization, and a configured engineering force threshold. The force-tightening coefficient is an empirical parameter of this model and task. When the constrained QP is infeasible or the configured delay threshold is exceeded, the fallback retains tip feedback but relaxes the hard RCM rows. The common-$P$ analysis applies to a separate six-state weighted state/control channel, not to nonlinear contact force or the deployed joint/task optimizer.

The current public evidence comprises the controller source, six focused regression tests, and the nominal, randomized, sensitivity, timing, and linear-design artifacts regenerated with the free-response fix and the retuned joint trust-region radius of Section VI-B both active (Sections VI–VI-G). Decision-coupled RCM authority remains the controller's clearest, order-of-magnitude benefit, and, with the trust region correctly tuned, joint/task MPC also matches or improves on tip-only MPC's tip tracking rather than trading it away. The RCM-band sweep and the trocar-clearance test still change character relative to the pre-correction controller (a non-monotonic band response and a real, if modest, clearance sensitivity, respectively, where the pre-correction measurements had shown a smoother trend and negligible sensitivity); these two sweeps are unaffected by the trust-region value (Section VI), so that shift is attributable to the free-response fix. An ablation separately showed that an under-tuned trust-region value, not the free-response fix, was responsible for a worse-performing intermediate regeneration of the headline nominal-episode and randomized-episode numbers; Section III reports the retuned value and Section VI the ablation. Constraint-matched predictive-impedance comparisons, broader randomized coverage, nonlinear margins, ex vivo contact, and integrated-platform tests define distinct subsequent validation stages.

## VIII. Conclusion

This paper formulates robot-assisted laparoscopy as a predictive interaction-dynamics problem. The framework uses a constant-transition, configuration-dependent-input residual model and a joint/task optimizer to represent tool-tip motion, linearized tool–trocar RCM consistency, total actuator authority, and adaptive tool–tissue penetration in one decision. The implementation includes guarded tissue estimation, a periodic free-response option, and an explicit feedback fallback; focused tests verify that the corrected free response reaches the deployed QP and influences its command and force rows.

The accompanying evaluation protocol separates the deployed nonlinear controller from the six-state linear design study and separates solver timing from complete controller-call timing. With the free-response fix active and the joint trust-region radius of Section III retuned from an initially too-tight 10 mrad to 20 mrad, the complete joint/task controller reduces nominal maximum RCM deviation to 0.59 mm and tip RMSE to 1.71 mm, against 5.53 mm RCM deviation and 2.55 mm RMSE for tip-only MPC and 8.13 mm RCM deviation for position impedance, while limiting a deep-reference force test to 1.46 N under a 3 N threshold; across eight randomized episodes it gives $2.49\pm0.93$ mm tip RMSE and $0.62\pm0.22$ mm maximum RCM deviation, with QP-solve fallbacks in six of eight episodes. The mistuned trust-region value, not the free-response fix, was responsible for an earlier regeneration's worse numbers (2.61 mm RMSE, 0.82 mm RCM deviation on the same nominal episode), identified by an ablation reported in Section VI. The evidence supports decision-coupled RCM authority as the controller's distinguishing mechanism. Within the stated local-model assumptions, the same formulation can be studied for other constrained contact tasks by replacing the task kinematics, contact output, and validated operating envelope.

## References


[1] Y. Cao and J. Tang, "Toward Interaction Dynamics: A Predictive Framework for Safe Physical Human–Robot Interaction," arXiv:2606.08281, Jun. 2026.

[2] J. Xue, W. Liang, Y. Wu, and T. H. Lee, "Model predictive variable impedance control towards safe robotic interaction in unknown disturbance-rich environments," *Robot. Auton. Syst.*, vol. 189, p. 104961, 2025.

[3] A. S. Anand *et al.*, "Deep model predictive variable impedance control," arXiv:2209.09614, 2022.

[4] D. Larby and F. Forni, "A passivity preserving $H_\infty$ synthesis technique for robot control," in *Proc. IEEE Conf. Decis. Control (CDC)*, 2022.

[5] G. Pannocchia and J. B. Rawlings, "Disturbance models for offset-free model-predictive control," *AIChE J.*, vol. 49, no. 2, pp. 426–437, Feb. 2003.

[6] Y.-Y. Cao, Z. Lin, and D. G. Ward, "Anti-windup design of output tracking systems subject to actuator saturation and constant disturbances," *Automatica*, vol. 40, no. 7, pp. 1221–1228, Jul. 2004.

[7] N. Hogan, "Impedance control: An approach to manipulation, Parts I–III," *ASME J. Dyn. Syst. Meas. Control*, vol. 107, no. 1, pp. 1–24, Mar. 1985.

[8] O. Khatib, "A unified approach for motion and force control of robot manipulators: The operational space formulation," *IEEE J. Robot. Autom.*, vol. 3, no. 1, pp. 43–53, Feb. 1987.

[9] K. H. Hunt and F. R. E. Crossley, "Coefficient of restitution interpreted as damping in vibroimpact," *ASME J. Appl. Mech.*, vol. 42, no. 2, pp. 440–445, Jun. 1975.

[10] N. Diolaiti, C. Melchiorri, and S. Stramigioli, "Contact impedance estimation for robotic systems," *IEEE Trans. Robot.*, vol. 21, no. 5, pp. 925–935, Oct. 2005.

[11] N. Aghakhani, M. Geravand, N. Shahriari, M. Vendittelli, and G. Oriolo, "Task control with remote center of motion constraint for minimally invasive robotic surgery," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2013, pp. 5807–5812.

[12] J. Sandoval, G. Poisson, and P. Vieyres, "A new kinematic formulation of the RCM constraint for redundant torque-controlled robots," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2017, pp. 4576–4581.

[13] D. Q. Mayne, J. B. Rawlings, C. V. Rao, and P. O. M. Scokaert, "Constrained model predictive control: Stability and optimality," *Automatica*, vol. 36, no. 6, pp. 789–814, Jun. 2000.

[14] D. Q. Mayne, M. M. Seron, and S. V. Raković, "Robust model predictive control of constrained linear systems with bounded disturbances," *Automatica*, vol. 41, no. 2, pp. 219–224, Feb. 2005.

[15] J. B. Rawlings, D. Q. Mayne, and M. Diehl, *Model Predictive Control: Theory, Computation, and Design*, 2nd ed. Madison, WI: Nob Hill Publishing, 2017.

[16] B. B. Peterson and K. S. Narendra, "Bounded error adaptive control," *IEEE Trans. Autom. Control*, vol. 27, no. 6, pp. 1161–1168, Dec. 1982.

[17] S. Boyd, L. El Ghaoui, E. Feron, and V. Balakrishnan, *Linear Matrix Inequalities in System and Control Theory*. SIAM, 1994.

[18] B. Stellato, G. Banjac, P. Goulart, A. Bemporad, and S. Boyd, "OSQP: An operator splitting solver for quadratic programs," *Math. Program. Comput.*, vol. 12, no. 4, pp. 637–672, 2020.